

Clustering Assignment Report

1. Application Goal

The objective of this assignment is to perform clustering on bank customer data to identify meaningful customer segments based on financial and economic indicators. The clusters can help banks tailor marketing strategies, recommend services, and better understand customer behavior.

2. Dataset Description

Dataset Name: bank-customers.csv

Source: Available publicly online (UCI repository / Kaggle)

This dataset includes client information from a Portuguese banking institution's marketing campaign. The original dataset contains 41,188 rows and several attributes related to customer demographics and economic indicators.

.Dataset Description - Column Overview:

- **age**: Age of the client (numeric)
- **job**: Type of job (e.g., admin., technician, unemployed)
- **marital**: Marital status (e.g., married, single)
- **education**: Education level (e.g., primary, secondary, tertiary)
- **default**: Has credit in default? (yes, no)
- **housing**: Has housing loan? (yes, no)
- **loan**: Has personal loan? (yes, no)
- **contact**: Contact communication type (cellular, telephone)
- **month**: Last contact month of the year
- **day_of_week**: Last contact day of the week
- **duration**: Last contact duration, in seconds (numeric)
- **campaign**: Number of contacts during this campaign
- **pdays**: Days since last contact from a previous campaign (-1 means never)
- **previous**: Number of contacts performed before this campaign
- **poutcome**: Outcome of the previous marketing campaign
- **emp.var.rate**: Employment variation rate (economic indicator)
- **cons.price.idx**: Consumer price index (monthly indicator)
- **cons.conf.idx**: Consumer confidence index (monthly indicator)
- **euribor3m**: Euribor 3-month rate (numeric, economic)
- **nr.employed**: Number of employees (economic indicator)
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3. Selected Features

The features selected for clustering are:

- emp.var.rate: Employment variation rate
- euribor3m: Euribor 3-month rate
- nr.employed: Number of employees
- cons.conf.idx: Consumer confidence index

These features were chosen after applying Exploratory Factor Analysis (EFA), where they showed high factor loadings and represent major economic drivers in customer behavior.

4. Cluster Tendency Test

To evaluate if the data has a tendency to form clusters, the Hopkins statistic was used.

Result: 1.0

Interpretation: A Hopkins score near 1 indicates highly clusterable data. However, a perfect score suggests potential redundancy or duplicated patterns in the dataset.

5. Determining Number of Clusters

To assess the optimal number of clusters (K), three methods were applied:

Calinski-Harabasz Index: Based on the score trend, the highest value occurred at K=10, confirming the previous methods.

- Elbow Method: Based on inertia, the plot showed an inflection point around K=10.
- Silhouette Score: Highest average silhouette score also occurred at K=10.

Conclusion: K = 10 was selected as the optimal number of clusters.

6. Clustering Algorithms Used

- KMeans clustering was applied with K=10.
- DBSCAN was applied with eps=0.5 and min_samples=15.
- Visualization of both methods was done using PCA-based scatter plots.

7. Evaluation of Clustering

Extrinsic evaluation was done using classification metrics with 'subscribed' as the ground truth label. Both algorithms were mapped to labels based on the majority class in each cluster.

KMeans Results:

- Accuracy: 89%
- Recall (subscribed): 0.20

DBSCAN Results:

- Accuracy: 89%
- Recall (subscribed): 0.19

Intrinsic evaluation used the Silhouette Score for both algorithms (excluding noise in DBSCAN).

8. Discussion

KMeans and DBSCAN performed similarly in terms of overall accuracy (~89%). However, KMeans offered clearer cluster boundaries and slightly higher recall for identifying the minority class (subscribed customers). DBSCAN was useful in detecting noise, but produced more unbalanced clusters, which slightly affected its recall. Therefore, KMeans is considered more suitable for this dataset due to:

- More stable and balanced cluster formation
- Higher interpretability of clusters
- Better recall for the target class
- Less sensitivity to noise in structured economic data